

1. A CCTV image contains salt and pepper noise

Median filter : It helps to remove salt and pepper noise, while preserving edges

2. A satellite image is corrupted by Gaussian noise
Gaussian filter helps to smooth this noise

3. A medical x-ray image has random noise but fine edge
Bilateral filter that removes noise while keeping fine edges sharp.

4. A scanned document has black and white dots
median filter is used to eliminate random black and white dots without affecting text

5. A low light image shows grain without boundaries
Non-local means filter is used to reduce grainy noise, while preserving the image details

6. A foggy image has very low contrast
Histogram Equalization that improves image contrast and visibility

7. A chest x-ray shows poor visibility of internal structure
CLAHE is used to enhance internal structure by improving local contrast

8. A night-time image lacks intensity range

9. Histogram equalization, it increases intensity variation and overall brightness

16. A gray scale image has only a narrow intensity range
contrast stretching Expand the intensity range to improve
contrast.

7. A under water image appears dull and washed out.
Laplace will enhance local contrast and improve visibility
without over-enhancement.

